RUBEN LORENZO

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EDUCATION

University of St. Thomas, St. Paul, MN Expected December 2026

Bachelor of Science in Mechanical Engineering, Minor in Peace Engineering

Honors: Dean's List (3x); UROP Research Grant (Summer 2024, Summer 2025); UST Student Leadership Scholarship (2024)

Relevant Coursework: Graduate-level: Advanced Thermal Systems, Advanced Product Design, Lean Six Sigma. Undergraduate: Fluid Dynamics, Thermodynamics/EES, System Dynamics & Controls (MATLAB/Simulink), Design of Experiments (Minitab)

EXPERIENCE

Sub Sea Systems, Inc., *Mechanical Engineering Intern* April 2026 – September 2026

- Diagnosed two failure modes (anti-rotation wear and off-axis binding) in the telescoping seat post of an underwater ride vehicle and led its SolidWorks redesign, delivering Rev B models and ANSI drawings
- Selected materials for constant chlorinated submersion: UHMW-PE bushings, 316/316L stainless steel and EPDM seals; ruled out acetal, oil-filled bronze and brass
- Sized and sourced a coaxial 316L gas strut with the supplier; documented 7 engineering changes in a revision log for review and change control
- Reverse-engineered existing components into SolidWorks models; produced Rev B drawings for sprockets and flanges and cut-length calculations for formed tubes

Minnesota Department of Transportation, *Engineering Intern, SEEDs Program* June 2025 – September 2026

- Collected and analyzed highway-corridor noise data with Class 1 sound level meters; applied the FHWA Traffic Noise Model (TNM) for compliance assessments and mapped receivers in ArcGIS Pro

ENGINEERING PROJECTS

Wind Tunnel Schlieren Module, Senior Design, *Project Manager (Sponsor: Twin City Fan)* June 2026 – December 2026

- Lead a 4-person team designing and building a bolt-on test section and Schlieren imaging system for TCF's AMCA 210 suction tunnel to sponsor requirements and a $10,000 budget; built the team's DFMEA and prepared design reviews
- Designed the inlet contraction and plywood diffuser; hand calculations set test-section flow at 55.3 m/s (Mach 0.16) with a 1.86 kPa pressure drop, under the 2.5 kPa limit
- Ran ANSYS Mechanical FEA with mesh refinement on test-section windows (1/4 vs. 3/8 in.); selected the high-speed camera, lens and mirrors

Hybrid Vertical-Axis Wind Turbine Study, *Independent Project (NPTV-funded)* 2026 – Present

- Extending a published hybrid VAWT study with upgrades to test new rotor combinations; built the SolidWorks model, scaled to 8.4% blockage for the UMN St. Anthony Falls Lab tunnel
- Developed the test method and fixture: servo holds fixed RPM, reaction torque sensor on a 3D-printed mount, and a no-wind tare at each RPM removes bearing friction; testing October 2026

Thermal Safety Analysis of a Wireless Neuromodulation Implant

- Built a transient bioheat transfer model in ANSYS CFX of implant recharging; identified 3.18 W as the maximum safe power input to prevent deep-tissue necrosis

RESEARCH

University of St. Thomas, Open-Roof Wind Tunnel Study (UROP) Summer – Fall 2025

- Designed a wind tunnel study comparing solid and 20% porous open-roof models at 10 and 20 m/s (Re ≈ 112,000–224,000)
- Designed and SLA-printed models with internal pressure-tap channels; porosity cut the average pressure differential by ~64% at 20 m/s
- Authored the technical paper and poster; sized models to a 7% blockage limit; ran 3+ trials per condition

University of St. Thomas, PFAS Wastewater Treatment Research June 2024 – May 2025

- Co-author: Ellis, Ling, Vermace, Lorenzo Quintana, Kyser, "Life-Cycle Environmental Impacts of Removing and Destroying PFAS From Wastewater Effluent," *Water Environment Research* (2026)

LEADERSHIP

Ruta'al, *Co-Founder & CEO* 2025 – 2026

- USSD-based financial services for unbanked rural users: 2nd place, Social Track, Fowler Global Social Innovation Challenge (Feb 2026); Global Finalist, San Diego (May 2026); $6,000 in prizes

University of St. Thomas, *Resident Advisor* August 2023 – May 2026

- Responsible for a floor of 30+ students: safety, policy, conflict resolution and after-hours emergency response

SKILLS

CAD & Simulation: SolidWorks (GD&T), ANSYS (Mechanical, Fluent, CFX), Fusion 360, MATLAB/Simulink, LabVIEW, EES, SimaPro
Quality & Statistics: DFMEA, Design of Experiments and statistical analysis (Minitab), Lean Six Sigma (graduate coursework)
Prototyping & Testing: CNC machining, welding, SLA/FDM 3D printing, fixtures, test method development
Languages: English and Spanish (bilingual)